

Response to the R65 Referee Report

Finite-Catalog Resource Allocation:
A Tariff-Sensitive Complexity Frontier

Anonymous revision R66 — September 27, 2026

We thank the referee for the careful independent audit and the recommendation to accept the paper. The report identifies no remaining scientific revision and asks that twelve qualifications be preserved during final preparation. We have retained each qualification and incorporated the optional clarification of what a verification receipt certifies. We have not changed a theorem, proof, mathematical display, empirical table, checker, request schema, or historical execution outcome. The complete R65 readers and response remain available in the preservation archive.

Editorial clarification of the receipt boundary

The companion now states explicitly that only mathematical conclusions represented in the returned verification receipt are certified, with their proof-class-specific meaning. A successful check does not endorse arbitrary additional metadata inside a low-level legacy certificate. The exact request and envelope schemas continue to reject unknown fields. The main article’s code-and-data paragraph provides a single-sentence statement of this boundary; implementation detail remains in the companion.

We also distinguish content binding from authenticated provenance. A digest does not establish the identity or authority of its supplier. The caller must obtain the external request from trusted storage or an authenticated channel; this remains an assumption outside the mathematical checker. Neither runtime attestation nor editorial approval can be acquired by adding such a claim to a certificate. This is an explanatory clarification of the existing interface, not a new checker contract or authentication layer.

Disposition of the twelve nonblocking comments

- 1. Physical regime.** The abstract, introduction and tariff section continue to specify common linear terminal reward and free preliminary service. The exact tariff theorem and proof are unchanged. Uniform fees are not claimed to solve the general curved-reward or costly-service model.
- 2. Near-standard tariffs.** We retain exact optimization of the sparse-exception surrogate, exact original-policy feasibility and fee evaluation, and an additive original-fee optimality certificate. No exact-tractability extension to arbitrary near-standard fees is introduced.
- 3. Projection class.** The abstract and projection subsection retain the specified sparse-exception class. Minimum distortion is not asserted over an unrestricted surrogate family. The attribution to trimmed absolute-deviation methods and the complete projection argument are unchanged.

4. **Exception counts.** The projection allowance, actual projected exception count and inner modal exception count remain distinct in the text, companion and projection-quality table. Their definitions, ordering and reported values are unchanged.
5. **Guard.** The companion continues to interpret a verified guard as agreement with the external mathematical request, not runtime attestation. Execution configuration and deadlines require separate immutable records.
6. **Evidence categories.** Mathematical validity, requested-width attainment, original on-time success and later integrity replay remain separate. A valid interval need not meet the requested tolerance. The historical failures and incomplete checks retain their original classification.
7. **Numerical solver evidence.** SCIP upper bounds remain numerical. Independently checked reconstructed policies certify lower bounds only. The clarification explicitly states that a lower-only receipt supplies no certified upper bound.
8. **Regret intervals.** Table 5 retains bracketed regret bounds wherever the comparator does not identify exact regret. No bounded regret is relabeled as an exact value, and the five identified zero regrets retain their exact comparator evidence.
9. **Historical wrappers.** The companion and request-contract document continue to disclose that wrappers are constructed at replay. Original certificate bytes are unchanged, and the new wrappers are not attributed to the original execution.
10. **Authentication.** Trusted storage and authenticated transport remain outside the mathematical certification scope. The new receipt paragraph makes this boundary explicit without asserting an unimplemented transport layer.
11. **Hybrid routing.** The computational section retains the tested hybrid and its mixed adversarial outcomes. It does not assert a universally optimal routing policy. All original requests and unsuccessful outcomes remain in the reported denominators.
12. **Expository placement.** The article remains centered on the selected-boundary representation and tariff-sensitive complexity frontier. Detailed certificate interpretation stays in the companion. No scientific content has been removed to make room for it.

Verification and preservation

The new receipt-scope regression uses existing frozen certificates from all six supported proof classes. It checks that adding unknown extension claims to legacy certificate bodies does not add certified conclusions to their returned receipts, while the corresponding unknown fields are rejected in the exact request and envelope. A nested tariff case and numerical lower-only semantics are also checked. The accepted checkers and their schemas are byte-identical to R65.

The package verifier compares every accepted mathematical environment, mathematical display, bibliography and table in the current readers with R65. It resolves all 7,080 original manifest entries through an explicit preservation map and requires unchanged existing code, inputs, proof files and execution records. New qualification reruns the request/transfer and retained semantic regressions, the structural suites and the complete frozen-proof replay. Original timing and target classifications are never overwritten by this replay. The exact commands, outcomes, file hashes and build checks accompany the revision.

A separate receipt binds the published commit and tree to the rebuilt readers and executed checks. It is evidence of artifact integrity and verification, not an editorial decision. This revision addresses the report's editorial comments without changing the scientific claims or checker contract that the referee found acceptable.